

LM Adapt Bench: LLM Continuous Adaptation & Forensic Contamination Audit

Language Model Adaptation Benchmarking Report

Date: 2026-07-07 04:22:58

Dataset: /leonardo_work/AIFAC_S03_029/fmihal00/
google_news_from_2026-06-08_to_2026-06-08_cleaned.jsonl

Model Count: 11

Search Integrity & Validation Summary

Range Health

Determines if our search space (e.g., Learning Rate range) was wide enough. **Optimal** means the model found its sweet spot. **Constrained** suggests a re-run with shifted ranges might yield better results.

Stability Index

Verifies if the short 200-step sweep accurately predicted long-term learning. An index > **90%** is a mathematical guarantee that our hyperparameters translated successfully to the full dataset.

Executive Summary

Rank	Model	Zero-shot BPB (Test)	Fine-tuned BPB (Test)	% Imp.	Stability	Convergence	Range Health	Time (min)
1	google/gemma-4-31B	0.753	0.583	22.6%	100.0%	Converged (Chained Training)	OPTIMAL	3983.0
2	google/gemma-4-12B	0.867	0.618	28.7%	98.0%	Converged (Chained Training)	OPTIMAL	1321.0
3	mistralai/Mistral-3-14B-Base-2512	0.723	0.651	9.9%	100.0%	Converged (Chained Training)	OPTIMAL	1012.0
4	Qwen/Qwen3.5-35B-A3B-Base	0.7	0.663	5.4%	97.0%	Converged (Chained Training)	OPTIMAL	2594.0
5	Qwen/Qwen3.5-9B-Base	0.733	0.679	7.4%	97.0%	Converged (Chained Training)	OPTIMAL	1518.0
6	Qwen/Qwen2.5-7B	0.753	0.728	3.3%	97.0%	Early Stopping (Converged)	OPTIMAL	198.0

Rank	Model	Zero-shot BPB (Test)	Fine-tuned BPB (Test)	% Imp.	Stability	Convergence	Range Health	Time (min)
7	Qwen/ Qwen3.5-4B-Base	0.775	0.757	2.3%	96.0%	Early Stopping (Converged)	OPTIMAL	280.0
8	meta-llama/ Llama-3.2-1B	0.814	0.773	5.1%	98.0%	Early Stopping (Converged)	OPTIMAL	78.0
9	Qwen/ Qwen2.5-1.5B	0.829	0.802	3.2%	99.0%	Early Stopping (Converged)	OPTIMAL	39.0
10	LiquidAI/ LFM2.5-1.2B-Base	1.346	0.854	36.5%	96.0%	Early Stopping (Converged)	OPTIMAL	66.0
11	Qwen/ Qwen2.5-0.5B	0.931	0.921	1.0%	99.0%	Early Stopping (Converged)	OPTIMAL	41.0

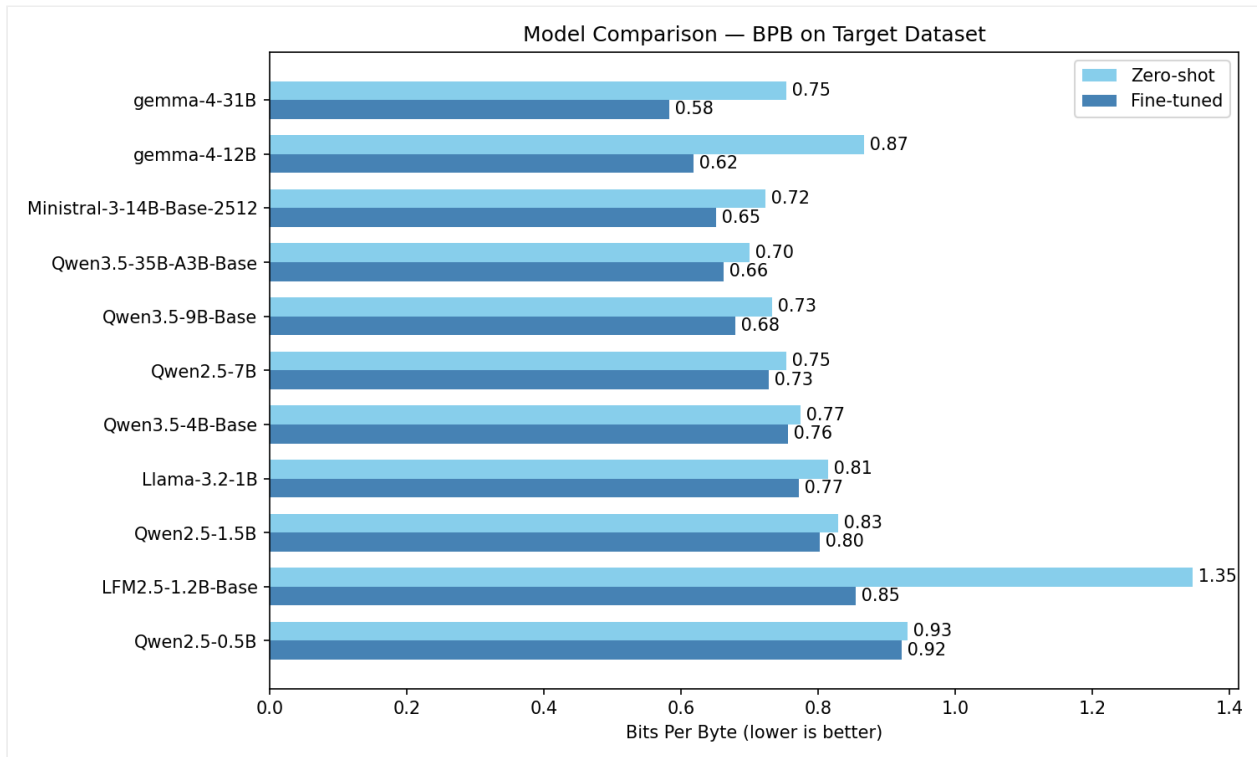


Figure 1: Comparison of zero-shot and fine-tuned BPB across all models.

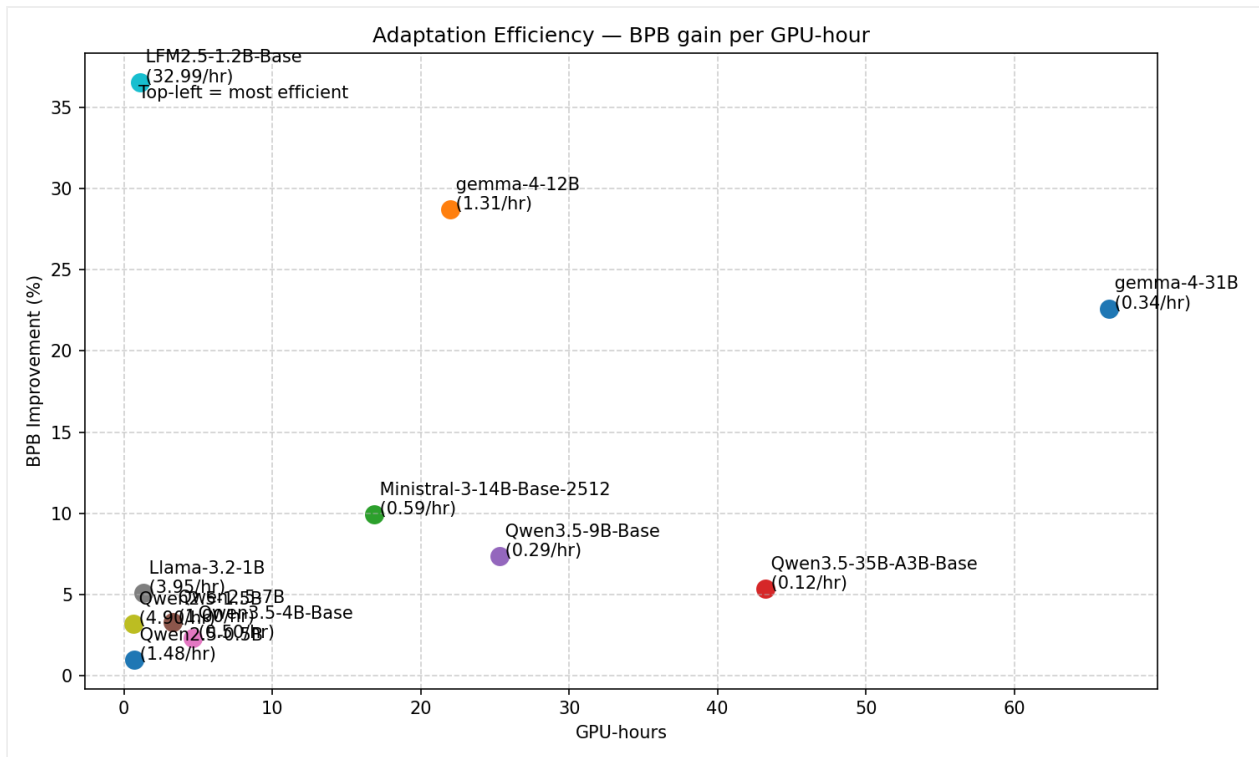


Figure 2: Adaptation efficiency showing improvement vs. compute cost. (Top-left is most efficient).

Hyperparameter Discovery Strategy

Each model underwent a 50-trial **Multi-Fidelity Search** using the Hyperband algorithm. This process automatically promotes promising configurations to higher data scales while pruning inefficient ones early.

Scientific Observations:

- **Categorical Robustness:** If a parameter (like *Batch Size*) shows "0" importance, it means the model family is robust. It achieved excellent results across all tested values, allowing you to prioritize the most memory-efficient setting (Batch Size 4) without losing quality.
- **Effective Throughput:** Small physical batches often win in LoRA adaptation because they perform more frequent updates per GPU-hour, leading to faster "conceptual discovery" in the target domain.

The Science of Scaling & Stability

A critical question in adaptation benchmarking is: *"Do hyperparameters found on a short run work for the full dataset?"*

Scaling Laws (Chinchilla): Our pipeline aligns learning dynamics between scales by proportionally stretching the **Cosine Learning Rate Decay**. This ensures that the exploration-to-exploitation trajectory is identical. Short sweep trials are not penalized by a lack of a cooldown phase, making hyperparameter performance at small scales a highly reliable predictor of full-scale adaptation behavior.

Stability Index Validation: Each model is assigned a **Stability Index**. We measure the Bits Per Byte (BPB) at exactly the end of the sweep budget in both the sweep and the final run. If these numbers match (Index > 95%), we have statistical evidence that our optimization parameters scale predictably to the larger data regime.

Data Contamination Risk Assessment

This assessment provides a multi-signal audit of the benchmark dataset's integrity, designed to detect and mitigate pre-training data memorization. Our framework combines lexical hygiene (SHA256/MinHash) with model-specific forensics (Min-K++/CoDeC), utilizing statistical outlier detection to ensure performance gains represent genuine learning rather than rote recall. This integrated approach provides a defensible evaluation of model adaptation validity by cross-referencing hard leakage with deep likelihood analysis.

OVERALL DATASET RISK

CRITICAL

AUDIT TIMESTAMP
2026-06-22T19:17:51.394549

Conclusion: Significant contamination or leakage detected. Benchmark integrity is compromised; evaluation splits were cleaned per policy.

Disclaimer: Residual risk remains because model pretraining corpora are not fully observable. These signals are statistical estimates and do not constitute absolute proof of absence or presence of contamination.

Dataset Hygiene Summary

Metric	Value	Description
Original Samples	119054	Total records loaded from source.
Duplicates Removed	10	Exact or near-duplicates found in the source.
Split Leakage	5	Cross-contamination between Train/Val/Test.
Forbidden Overlap	0	Matches found in configured public corpora.
Lexical Quarantined / Dropped	15	Total lexical samples removed globally from Train/Val/Test.
Global Forensic Drops (Unified Set)	80	Unified forensic outliers quarantined across all models.
Cleaned Final Count	119040	Total samples used in the benchmark training.

Per-Model Contamination Signal Matrix

Model ID	Split	Variant	Risk Label	CoDeC Outliers	Min-K++ Outliers	Forensic Drops	DCQ Confirmation*
google/gemma-4-31B	train	Min-K+ +	MEDIUM	0.0%	3.5%	16	100.0%
google/gemma-4-31B	val	Min-K+ +	MEDIUM	0.2%	3.2%	19	100.0%
google/gemma-4-31B	test	Min-K+ +	MEDIUM	0.6%	3.1%	20	100.0%
google/gemma-4-12B	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
google/gemma-4-12B	val	Min-K+ +	MEDIUM	0.4%	4.0%	11	100.0%
google/gemma-4-12B	test	Min-K+ +	MEDIUM	0.6%	4.0%	11	100.0%
mistralai/ Ministral-3-14B- Base-2512	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
mistralai/ Ministral-3-14B- Base-2512	val	Min-K+ +	MEDIUM	0.0%	3.8%	13	100.0%
mistralai/ Ministral-3-14B- Base-2512	test	Min-K+ +	MEDIUM	0.8%	3.8%	13	100.0%
Qwen/Qwen3.5-35B- A3B-Base	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen3.5-35B- A3B-Base	val	Min-K+ +	MEDIUM	0.0%	3.4%	17	100.0%
Qwen/Qwen3.5-35B- A3B-Base	test	Min-K+ +	MEDIUM	0.8%	4.0%	11	100.0%
Qwen/Qwen3.5-9B- Base	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen3.5-9B- Base	val	Min-K+ +	MEDIUM	0.2%	3.0%	21	100.0%
Qwen/Qwen3.5-9B- Base	test	Min-K+ +	MEDIUM	0.6%	3.4%	17	100.0%

Model ID	Split	Variant	Risk Label	CoDeC Outliers	Min-K++ Outliers	Forensic Drops	DCQ Confirmation*
Qwen/Qwen2.5-7B	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen2.5-7B	val	Min-K+ +	MEDIUM	0.2%	3.8%	13	100.0%
Qwen/Qwen2.5-7B	test	Min-K+ +	MEDIUM	0.6%	3.7%	14	100.0%
Qwen/Qwen3.5-4B-Base	train	Min-K+ +	MEDIUM	0.0%	3.6%	15	100.0%
Qwen/Qwen3.5-4B-Base	val	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen3.5-4B-Base	test	Min-K+ +	MEDIUM	0.6%	3.9%	12	100.0%
meta-llama/Llama-3.2-1B	train	Min-K+ +	MEDIUM	0.0%	2.7%	24	100.0%
meta-llama/Llama-3.2-1B	val	Min-K+ +	MEDIUM	0.0%	2.5%	25	96.2%
meta-llama/Llama-3.2-1B	test	Min-K+ +	MEDIUM	0.0%	2.3%	28	100.0%
Qwen/Qwen2.5-1.5B	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen2.5-1.5B	val	Min-K+ +	MEDIUM	0.4%	3.7%	14	100.0%
Qwen/Qwen2.5-1.5B	test	Min-K+ +	MEDIUM	0.4%	3.4%	17	100.0%
LiquidAI/LFM2.5-1.2B-Base	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
LiquidAI/LFM2.5-1.2B-Base	val	Min-K+ +	MEDIUM	0.2%	3.6%	15	100.0%
LiquidAI/LFM2.5-1.2B-Base	test	Min-K+ +	MEDIUM	0.8%	2.8%	22	95.7%
Qwen/Qwen2.5-0.5B	train	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%
Qwen/Qwen2.5-0.5B	val	Min-K+ +	MEDIUM	0.0%	4.0%	11	100.0%

Model ID	Split	Variant	Risk Label	CoDeC Outliers	Min-K++ Outliers	Forensic Drops	DCQ Confirmation*
Qwen/Qwen2.5-0.5B	test	Min-K++	MEDIUM	0.2%	3.8%	13	100.0%

Note on Model-Based Risk Signals: These are statistical estimates and do not constitute absolute proof. **Min-K++** is a sample-level familiarity signal that evaluates whether difficult tokens are unusually likely relative to the model's predicted vocabulary distribution. **Min-K-style** is a fallback that averages raw next-token log probabilities. **CoDeC** is a split-level signal that detects if adding context makes the model suspiciously more confident. **DCQ** (when run) is a confirmation check using perturbed alternatives. ***DCQ Confirmation Rate:** This represents the percentage of highly-suspicious outliers (Min-K++ candidates) that failed the forced-choice quiz, confirming they are memorized leaks. It is a measure of forensic accuracy on a filtered subset, **not** the global contamination rate of the entire dataset.

Metric Glossary

Bits Per Byte (BPB) is the core metric used in this benchmark. It measures how effectively a language model compresses a given dataset. Unlike perplexity, BPB is **tokeniser-agnostic**, meaning it can be used to compare models with different vocabularies fairly.

Effective Batch Size is calculated as `Batch Size * Gradient Accumulation Steps`. High effective batch sizes often lead to more stable gradients and better convergence on complex datasets, even when physical VRAM is limited.

Formula: $BPB = (\text{Mean Cross-Entropy Nats} / \ln(2)) / \text{Average Bytes Per Token}$

Lower BPB values indicate better adaptation to the target domain. A reduction in BPB after fine-tuning represents successful learning of the dataset's patterns.

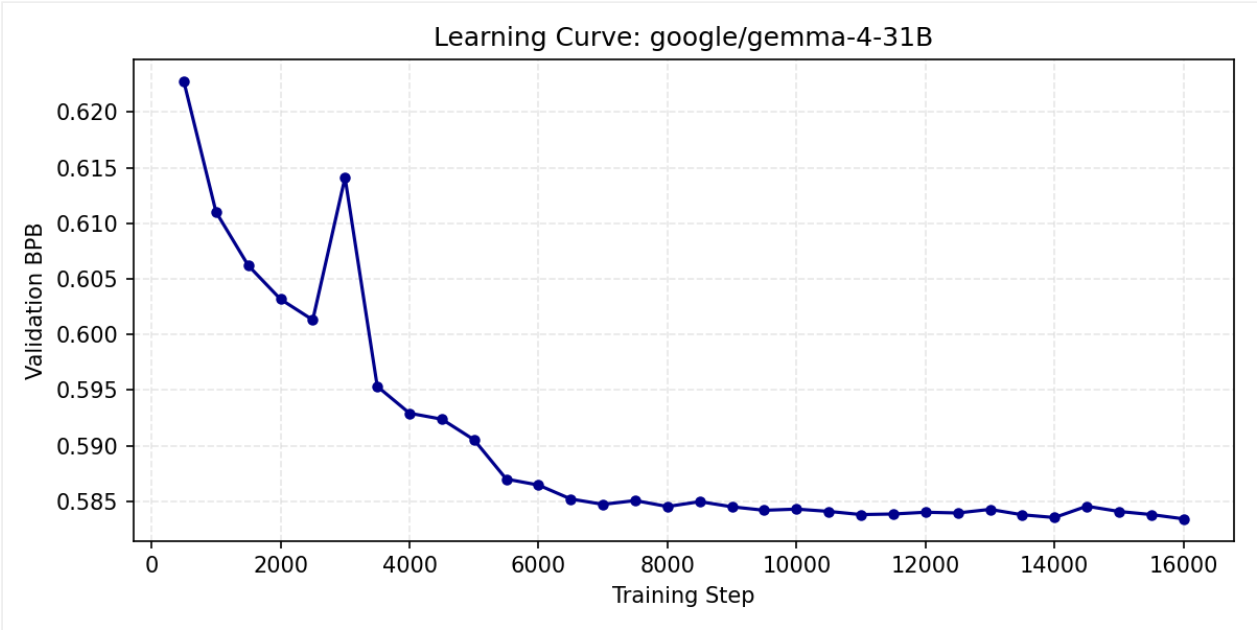
google/gemma-4-31B

TOTAL PARAMS
31396.0M
TRAINABLE %
0.39%
ZERO-SHOT BPB
0.753
FINE-TUNED BPB
0.583
EFFECTIVE BATCH
16
STABILITY INDEX
99.6%

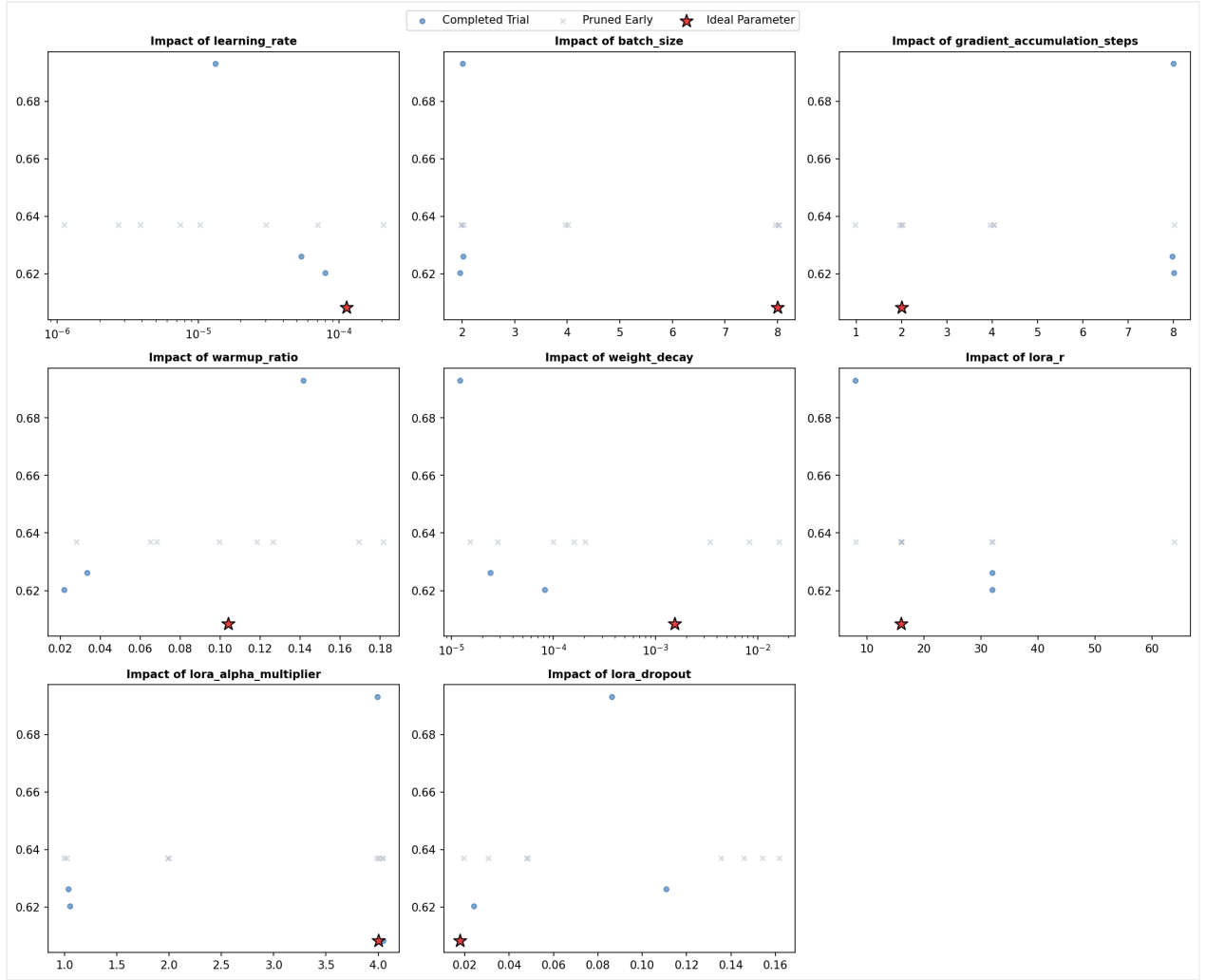
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011290133559092664	OPTIMAL
batch_size	8	OPTIMAL
warmup_ratio	0.10401360423556216	OPTIMAL
weight_decay	0.0015375920235481753	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.0176985004103839	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED
max_train_seconds	600.0	FIXED
target_train_steps	8000	FIXED
lr_floor_ratio	0.1	FIXED



Learning curve for google/gemma-4-31B.



Hyperparameter sweep results for google/gemma-4-31B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

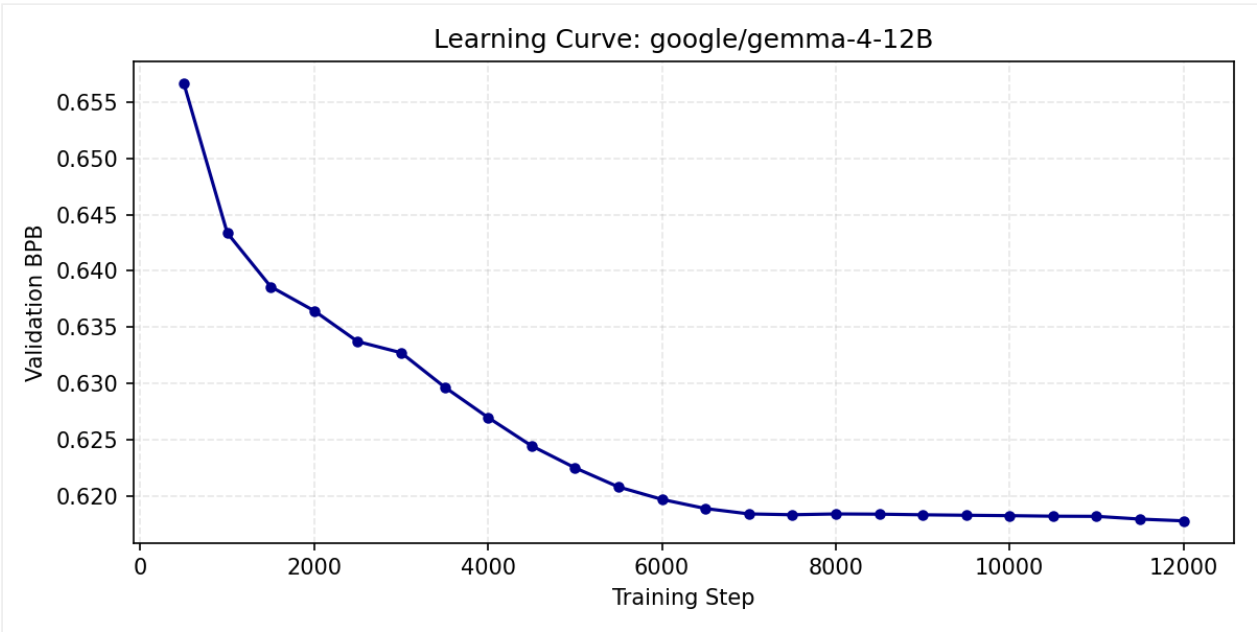
google/gemma-4-12B

TOTAL PARAMS
12025.0M
TRAINABLE %
0.55%
ZERO-SHOT BPB
0.867
FINE-TUNED BPB
0.618
EFFECTIVE BATCH
16
STABILITY INDEX
98.3%

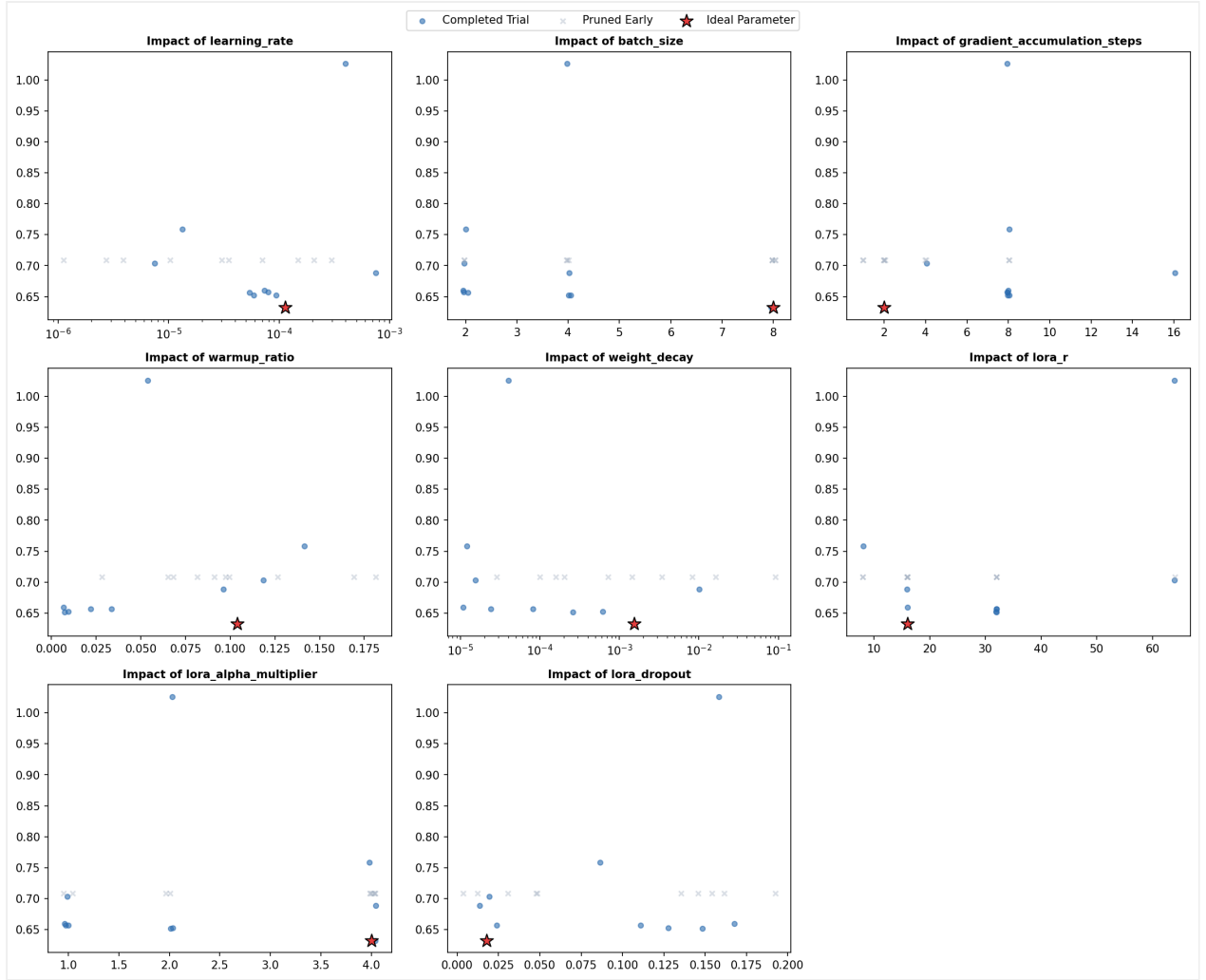
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011290133559092664	OPTIMAL
batch_size	8	OPTIMAL
warmup_ratio	0.10401360423556216	OPTIMAL
weight_decay	0.0015375920235481753	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.0176985004103839	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED
max_train_seconds	74137.46951389313	FIXED
target_train_steps	8000	FIXED
lr_floor_ratio	0.1	FIXED



Learning curve for google/gemma-4-12B.



Hyperparameter sweep results for google/gemma-4-12B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

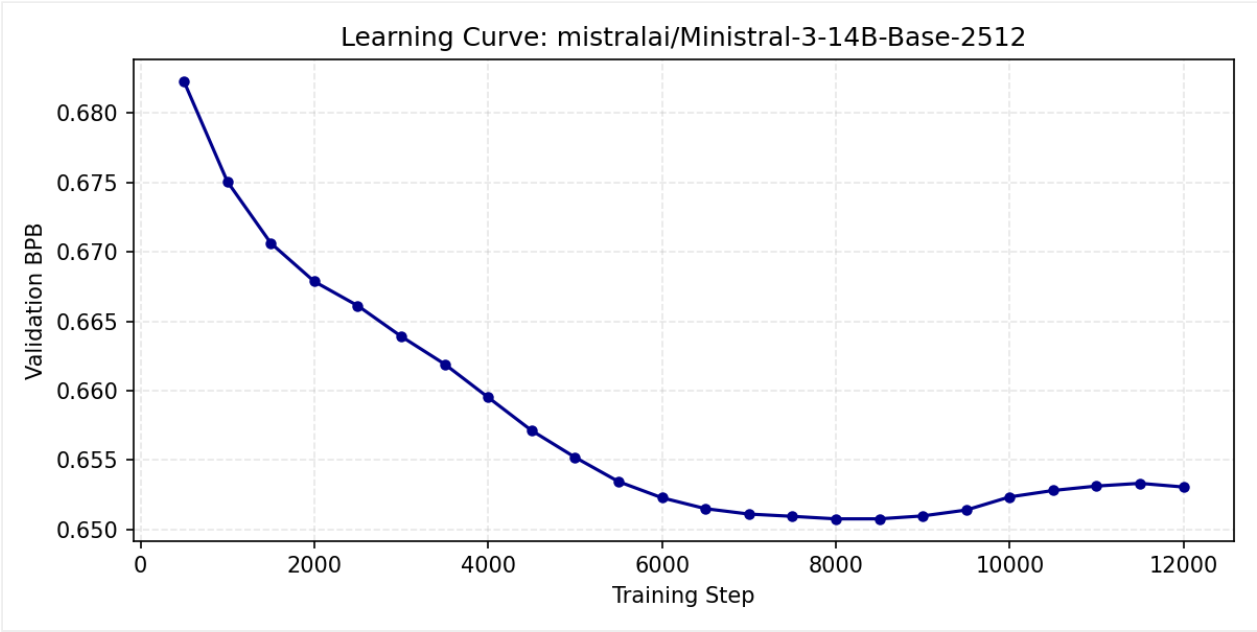
mistralai/Ministral-3-14B-Base-2512

TOTAL PARAMS
14006.0M
TRAINABLE %
0.44%
ZERO-SHOT BPB
0.723
FINE-TUNED BPB
0.651
EFFECTIVE BATCH
16
STABILITY INDEX
99.7%

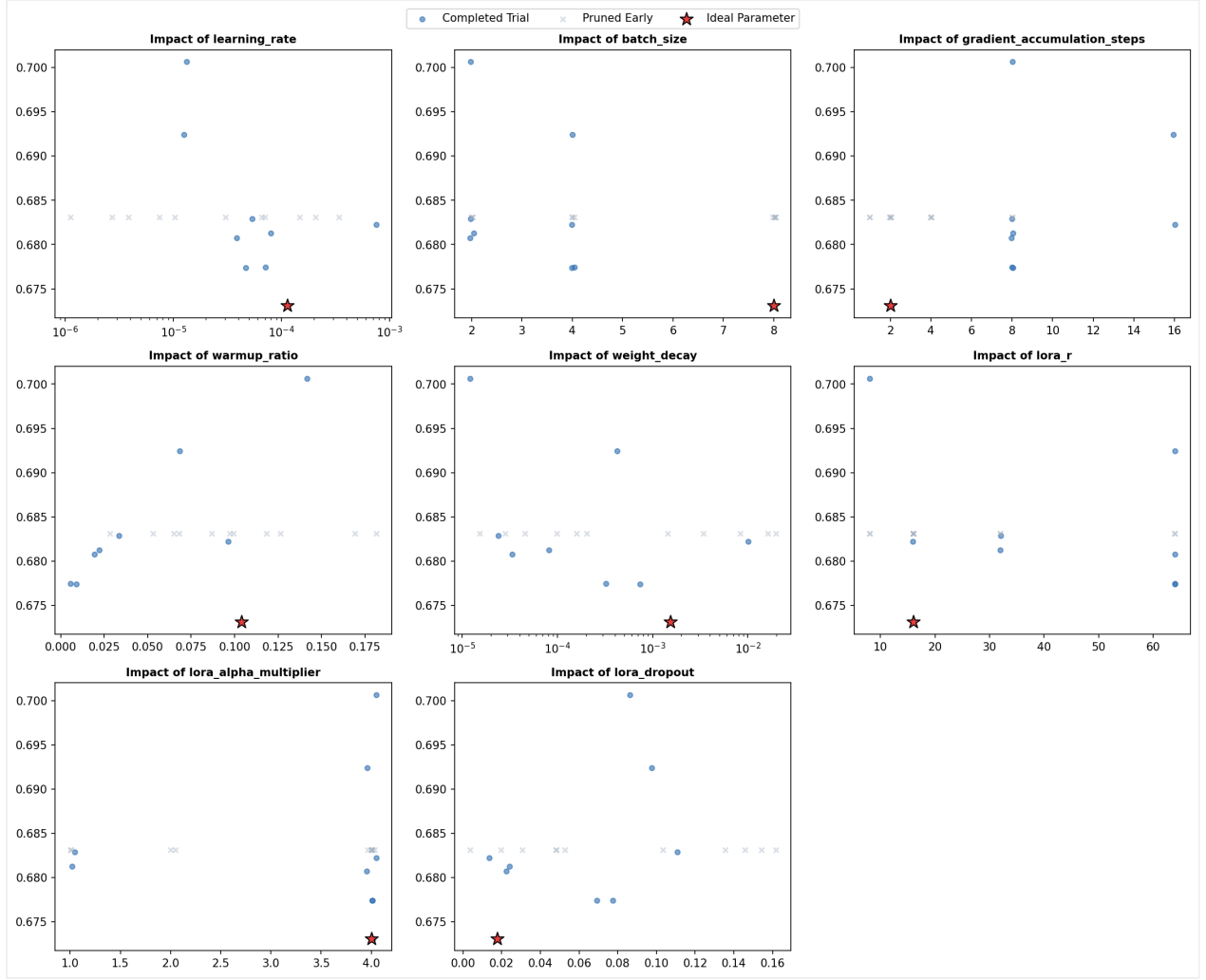
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011290133559092664	OPTIMAL
batch_size	8	OPTIMAL
warmup_ratio	0.10401360423556216	OPTIMAL
weight_decay	0.0015375920235481753	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.0176985004103839	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED
max_train_seconds	74693.8721408844	FIXED
target_train_steps	8000	FIXED
lr_floor_ratio	0.1	FIXED



Learning curve for mistralai/Ministral-3-14B-Base-2512.



Hyperparameter sweep results for mistralai/Mistral-3-14B-Base-2512. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

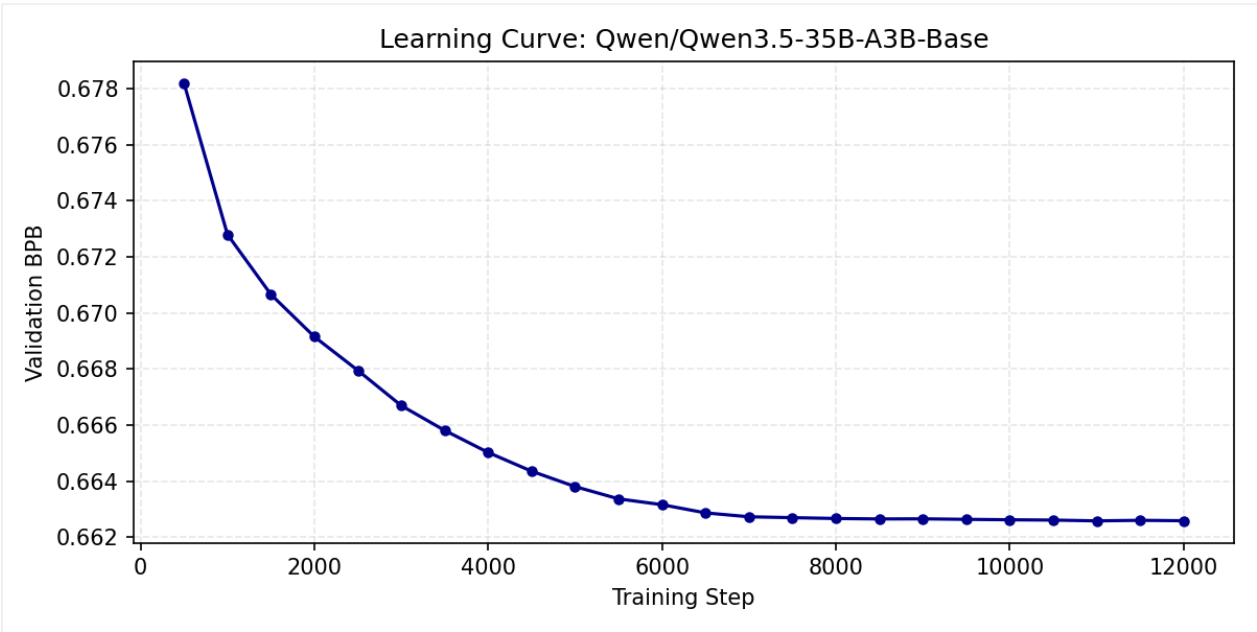
Qwen/Qwen3.5-35B-A3B-Base

TOTAL PARAMS
34669.0M
TRAINABLE %
0.02%
ZERO-SHOT BPB
0.7
FINE-TUNED BPB
0.663
EFFECTIVE BATCH
16
STABILITY INDEX
97.2%

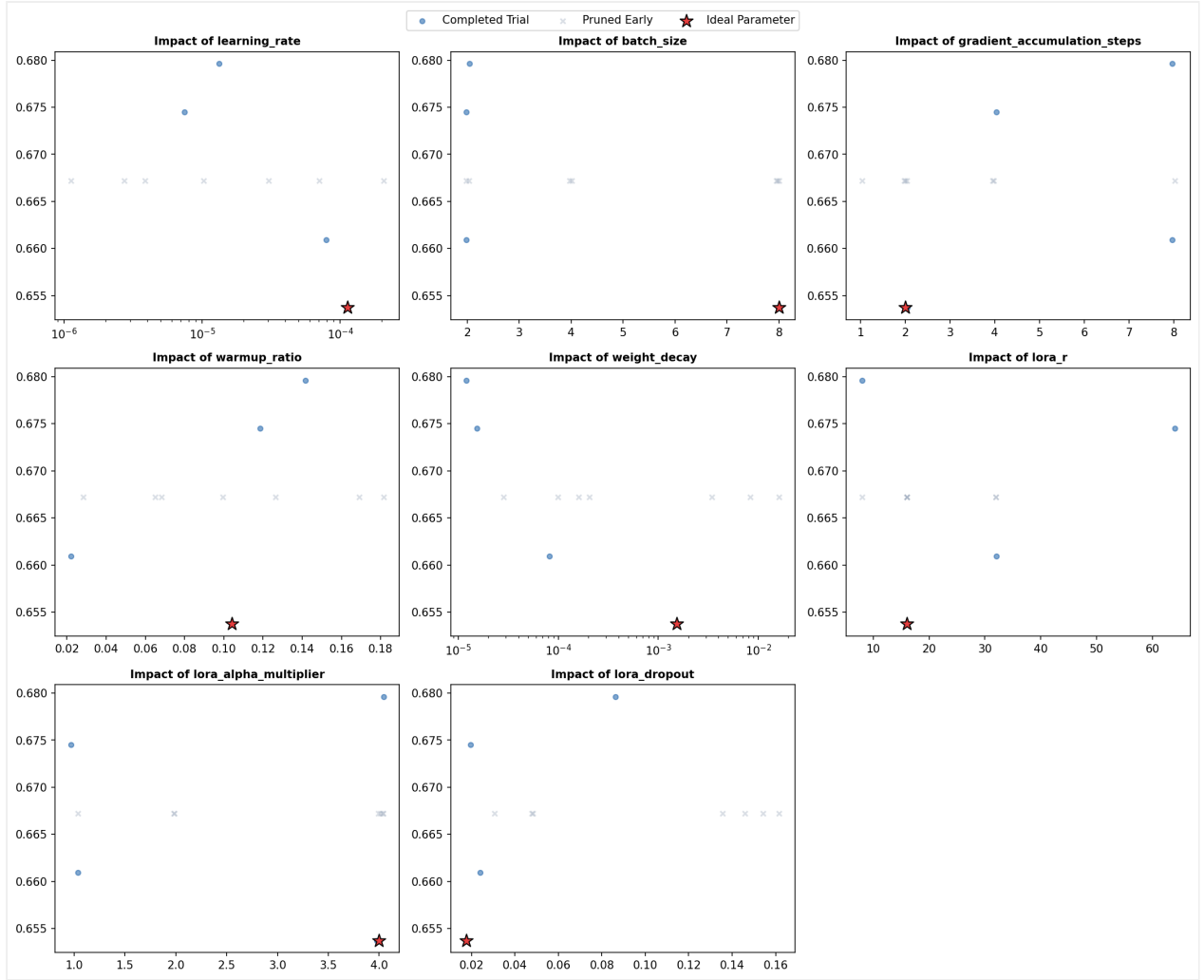
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011290133559092664	OPTIMAL
batch_size	8	OPTIMAL
warmup_ratio	0.10401360423556216	OPTIMAL
weight_decay	0.0015375920235481753	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.0176985004103839	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED
max_train_seconds	69830.70377612114	FIXED
target_train_steps	8000	FIXED
lr_floor_ratio	0.1	FIXED



Learning curve for Qwen/Qwen3.5-35B-A3B-Base.



Hyperparameter sweep results for Qwen/Qwen3.5-35B-A3B-Base. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

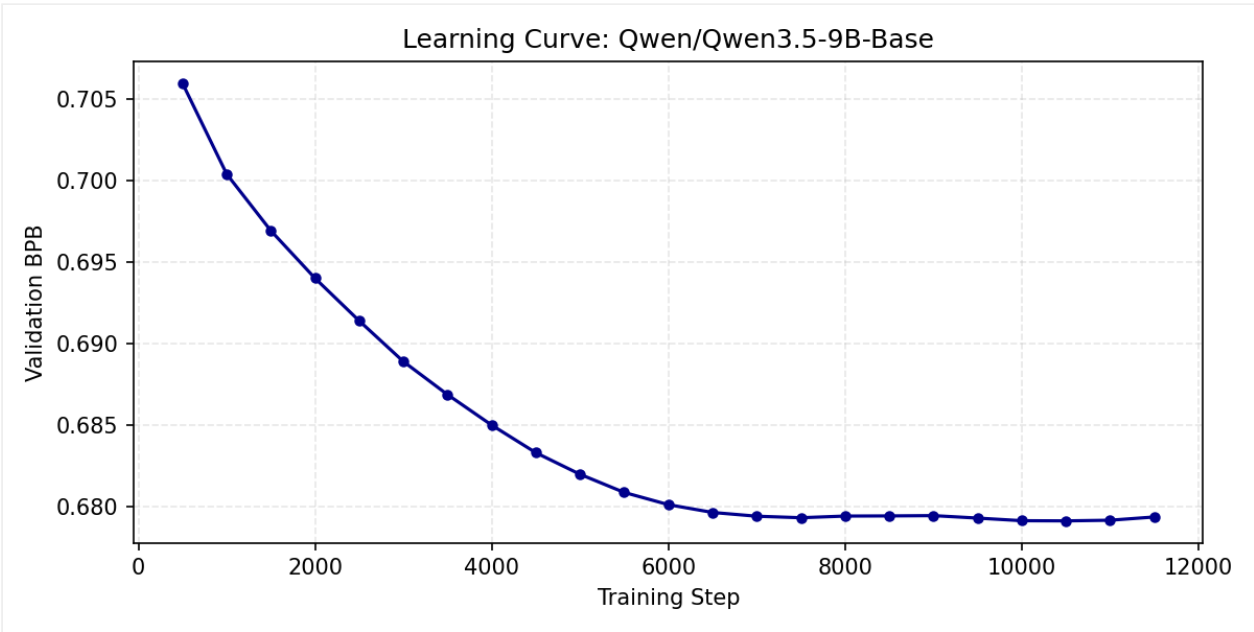
Qwen/Qwen3.5-9B-Base

TOTAL PARAMS
8983.0M
TRAINABLE %
0.32%
ZERO-SHOT BPB
0.733
FINE-TUNED BPB
0.679
EFFECTIVE BATCH
16
STABILITY INDEX
97.2%

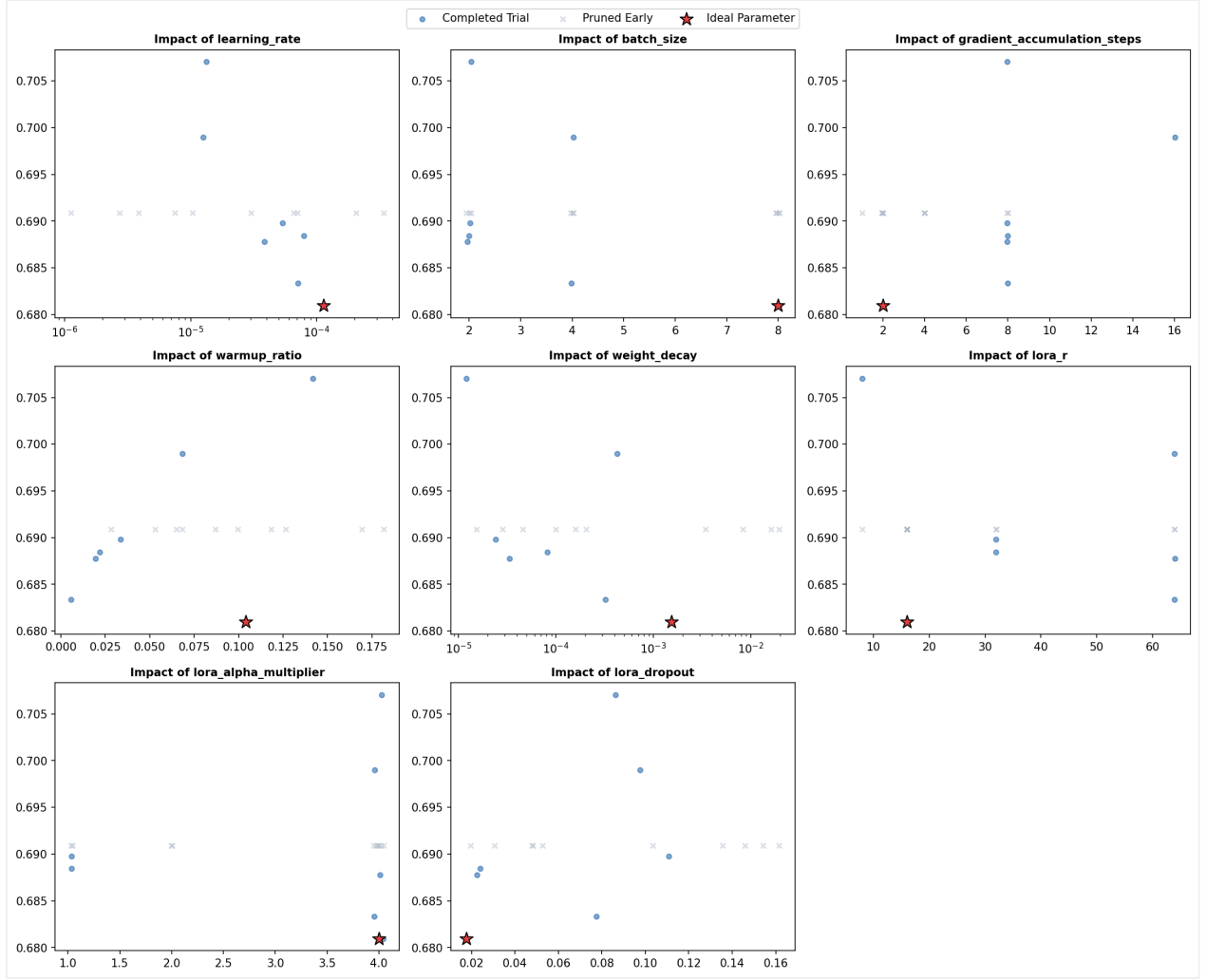
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011290133559092664	OPTIMAL
batch_size	8	OPTIMAL
warmup_ratio	0.10401360423556216	OPTIMAL
weight_decay	0.0015375920235481753	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.0176985004103839	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED
max_train_seconds	73460.07796764374	FIXED
target_train_steps	8000	FIXED
lr_floor_ratio	0.1	FIXED



Learning curve for Qwen/Qwen3.5-9B-Base.



Hyperparameter sweep results for Qwen/Qwen3.5-9B-Base. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

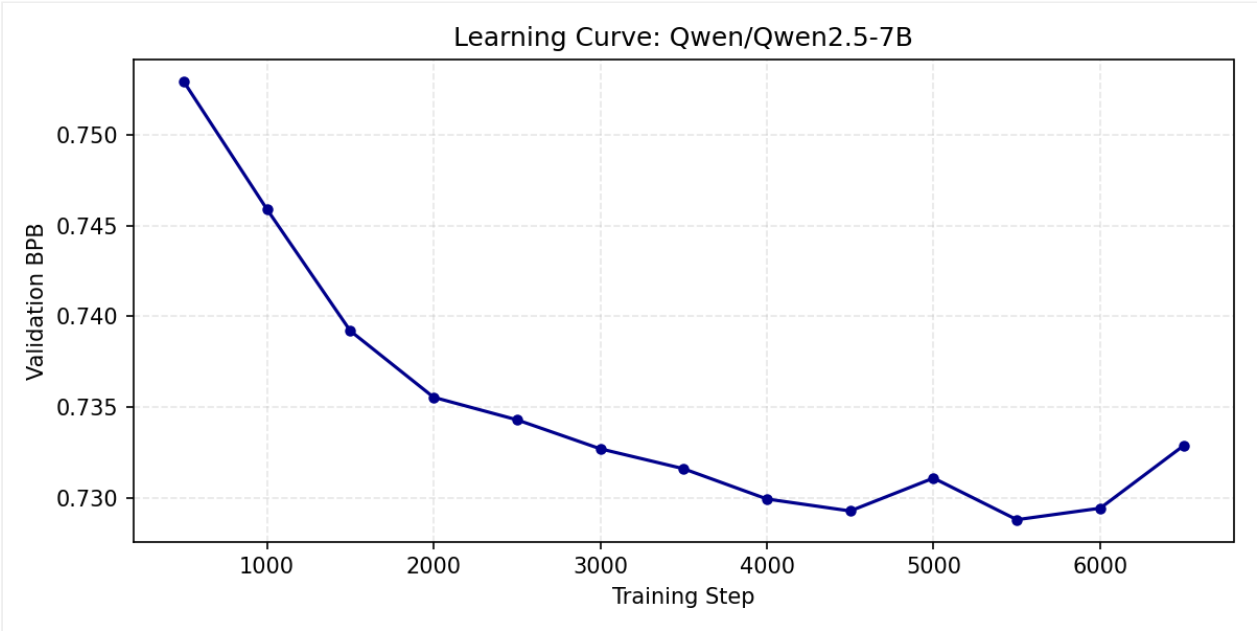
Qwen/Qwen2.5-7B

TOTAL PARAMS
7656.0M
TRAINABLE %
0.53%
ZERO-SHOT BPB
0.753
FINE-TUNED BPB
0.728
EFFECTIVE BATCH
2
STABILITY INDEX
97.0%

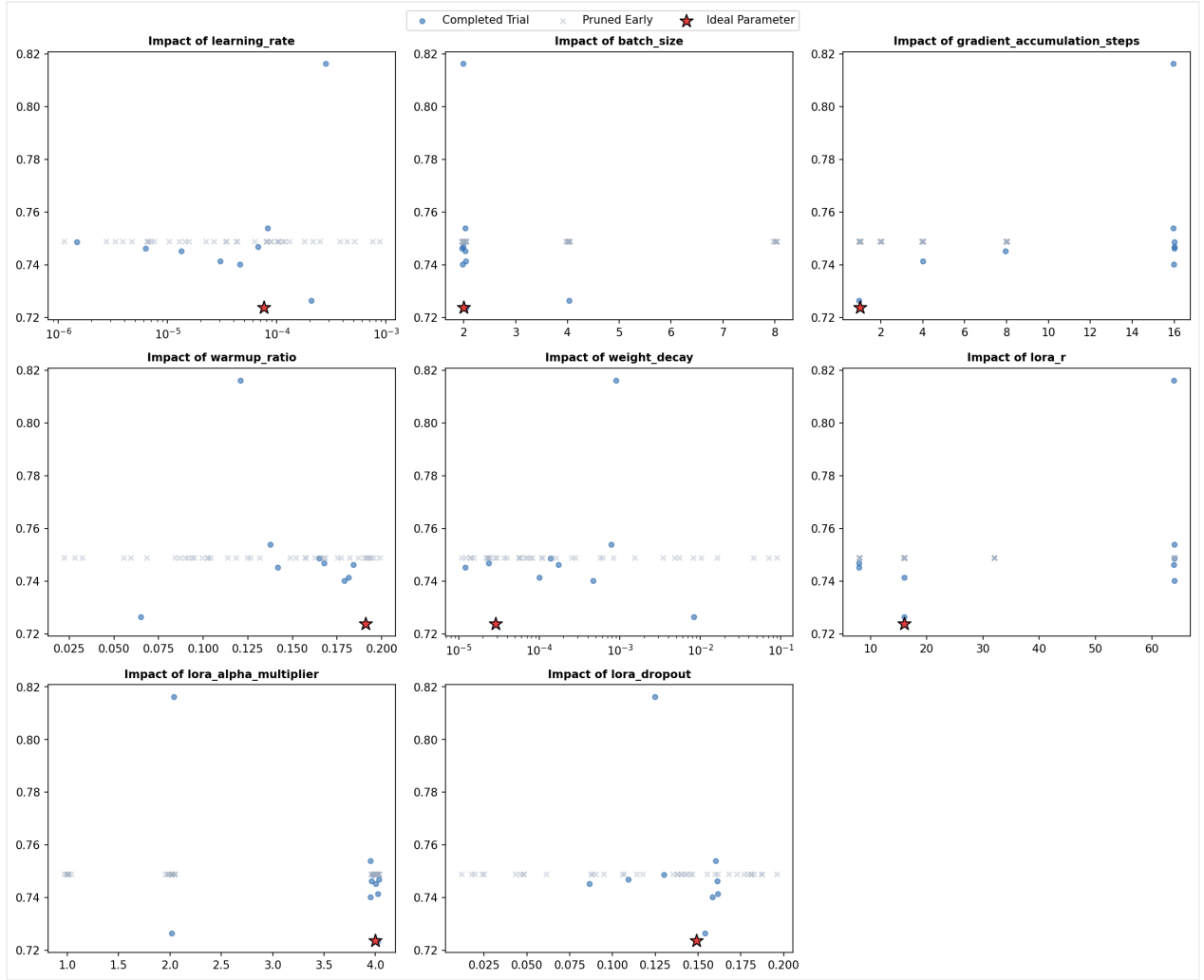
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	7.555232351885553e-05	OPTIMAL
batch_size	2	OPTIMAL
warmup_ratio	0.19104386880924357	CONSTRAINED
weight_decay	2.8762039921439462e-05	CONSTRAINED
gradient_accumulation_steps	1	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	64	FIXED
lora_dropout	0.14916524556038532	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for Qwen/Qwen2.5-7B.



Hyperparameter sweep results for Qwen/Qwen2.5-7B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

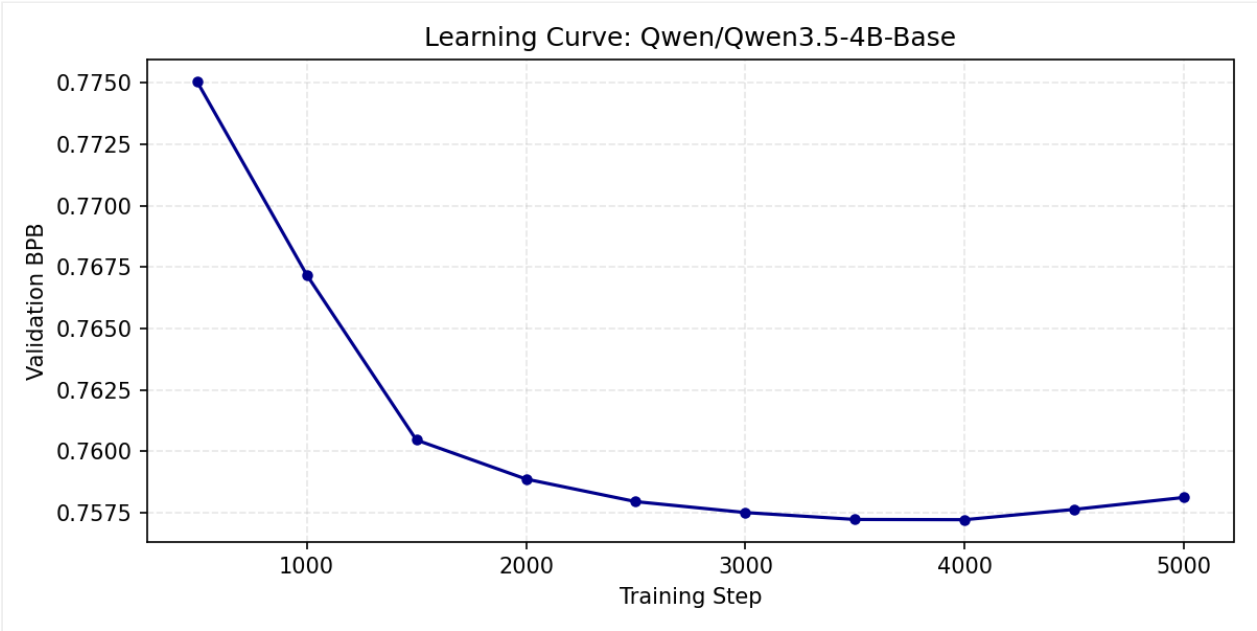
Qwen/Qwen3.5-4B-Base

TOTAL PARAMS
4216.0M
TRAINABLE %
0.25%
ZERO-SHOT BPB
0.775
FINE-TUNED BPB
0.757
EFFECTIVE BATCH
2
STABILITY INDEX
96.5%

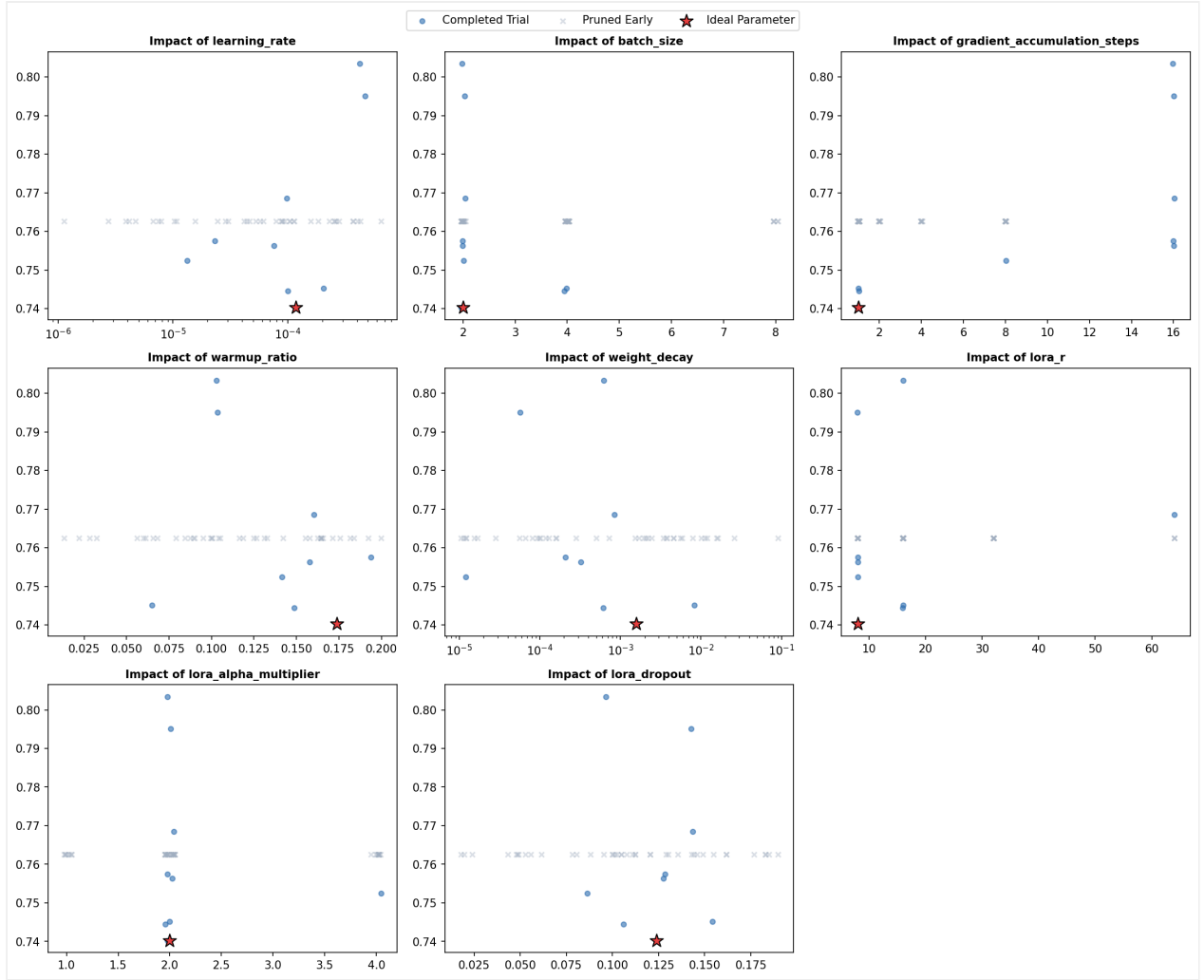
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00011795010004603094	OPTIMAL
batch_size	2	OPTIMAL
warmup_ratio	0.17395184560414692	OPTIMAL
weight_decay	0.0015712354148937214	CONSTRAINED
gradient_accumulation_steps	1	OPTIMAL
lora_r	8	OPTIMAL
lora_alpha	16	FIXED
lora_dropout	0.12395014198443678	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for Qwen/Qwen3.5-4B-Base.



Hyperparameter sweep results for Qwen/Qwen3.5-4B-Base. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

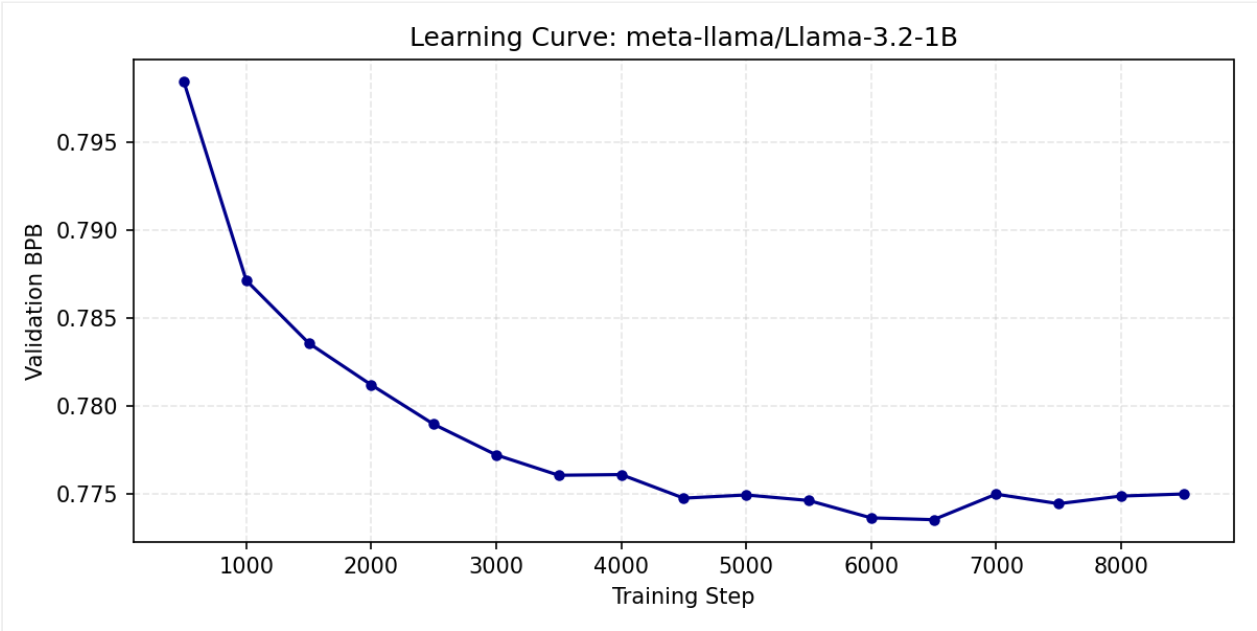
meta-llama/Llama-3.2-1B

TOTAL PARAMS
1247.0M
TRAINABLE %
0.9%
ZERO-SHOT BPB
0.814
FINE-TUNED BPB
0.773
EFFECTIVE BATCH
4
STABILITY INDEX
98.3%

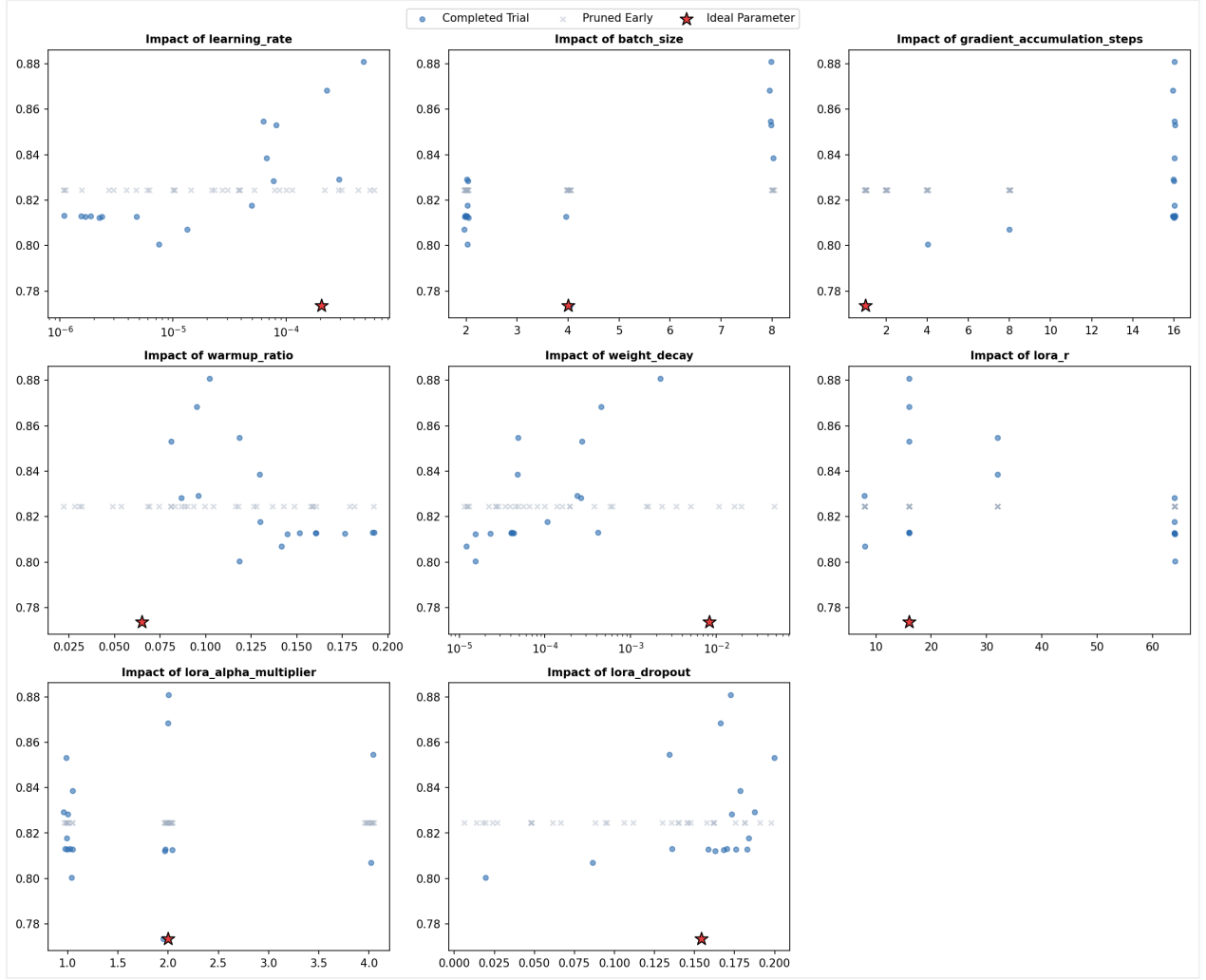
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00020597335357437193	OPTIMAL
batch_size	4	OPTIMAL
warmup_ratio	0.06503666440534941	OPTIMAL
weight_decay	0.008287522363768155	OPTIMAL
gradient_accumulation_steps	1	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	32	FIXED
lora_dropout	0.1541934359909122	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for meta-llama/Llama-3.2-1B.



Hyperparameter sweep results for meta-llama/Llama-3.2-1B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

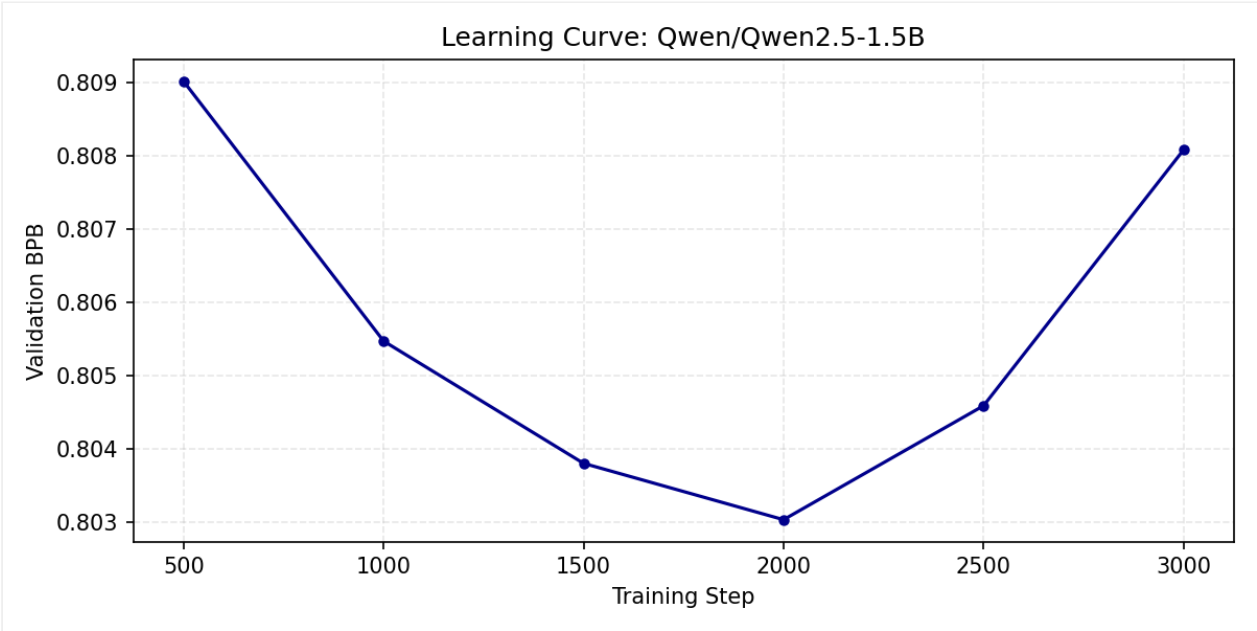
Qwen/Qwen2.5-1.5B

TOTAL PARAMS
1562.0M
TRAINABLE %
1.18%
ZERO-SHOT BPB
0.829
FINE-TUNED BPB
0.802
EFFECTIVE BATCH
4
STABILITY INDEX
98.9%

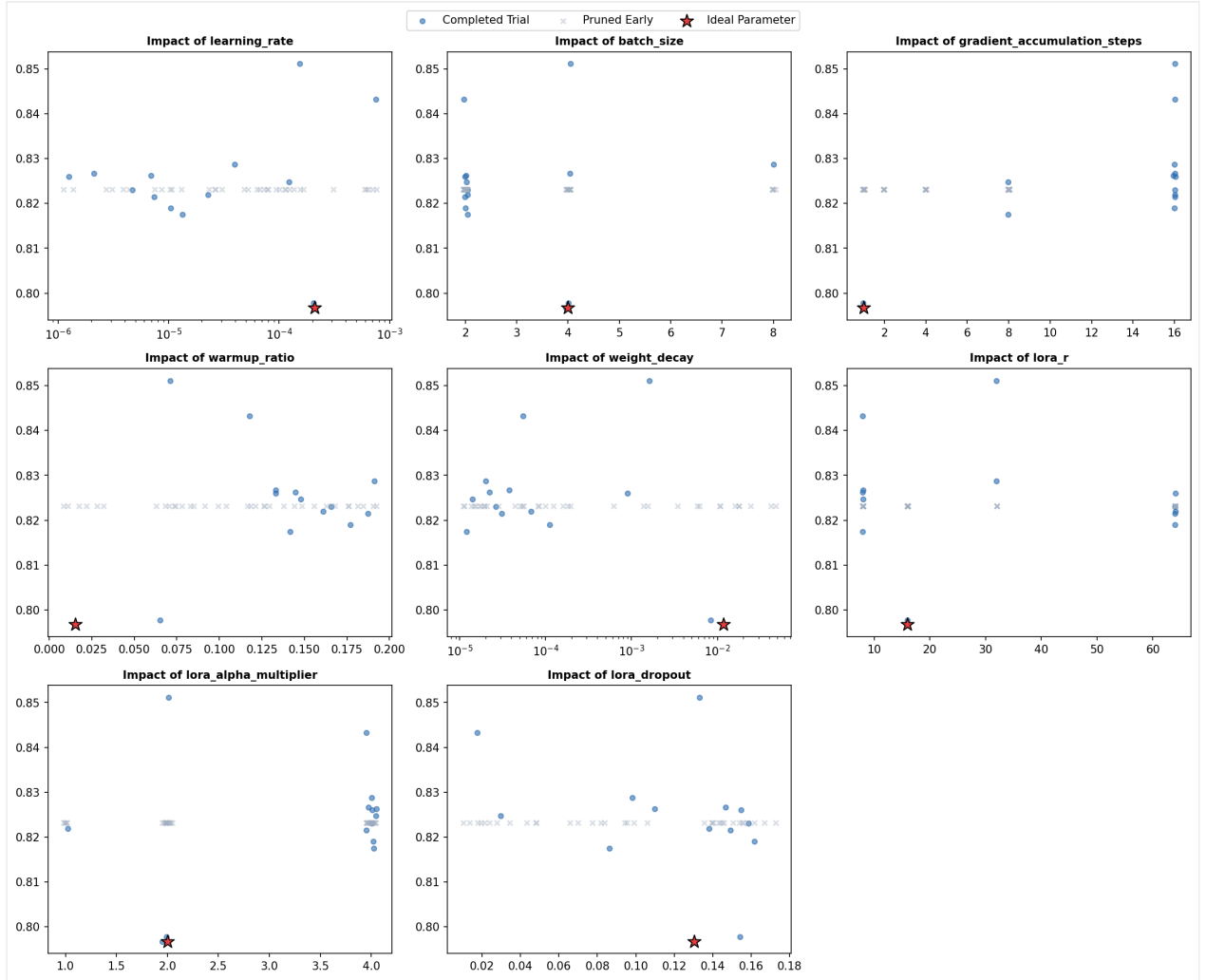
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.00020986449221531314	OPTIMAL
batch_size	4	OPTIMAL
warmup_ratio	0.015435621403515759	OPTIMAL
weight_decay	0.011717816987767596	OPTIMAL
gradient_accumulation_steps	1	OPTIMAL
lora_r	16	OPTIMAL
lora_alpha	32	FIXED
lora_dropout	0.1304031685263773	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for Qwen/Qwen2.5-1.5B.



Hyperparameter sweep results for Qwen/Qwen2.5-1.5B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

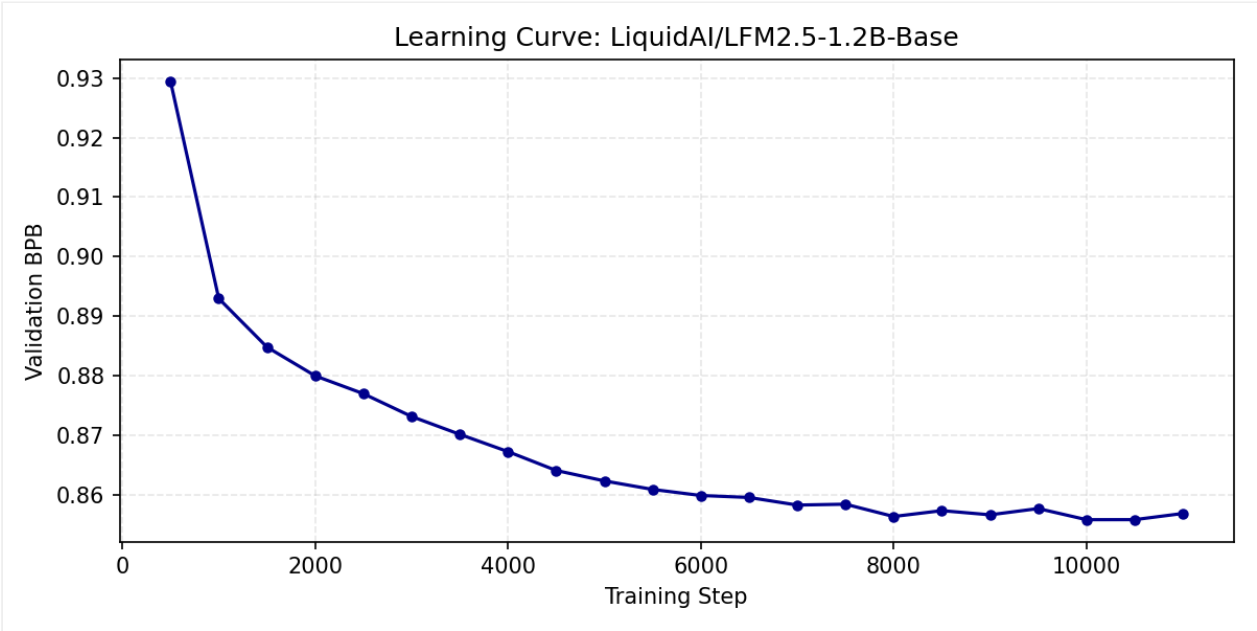
LiquidAI/LFM2.5-1.2B-Base

TOTAL PARAMS
1178.0M
TRAINABLE %
0.66%
ZERO-SHOT BPB
1.346
FINE-TUNED BPB
0.854
EFFECTIVE BATCH
2
STABILITY INDEX
96.2%

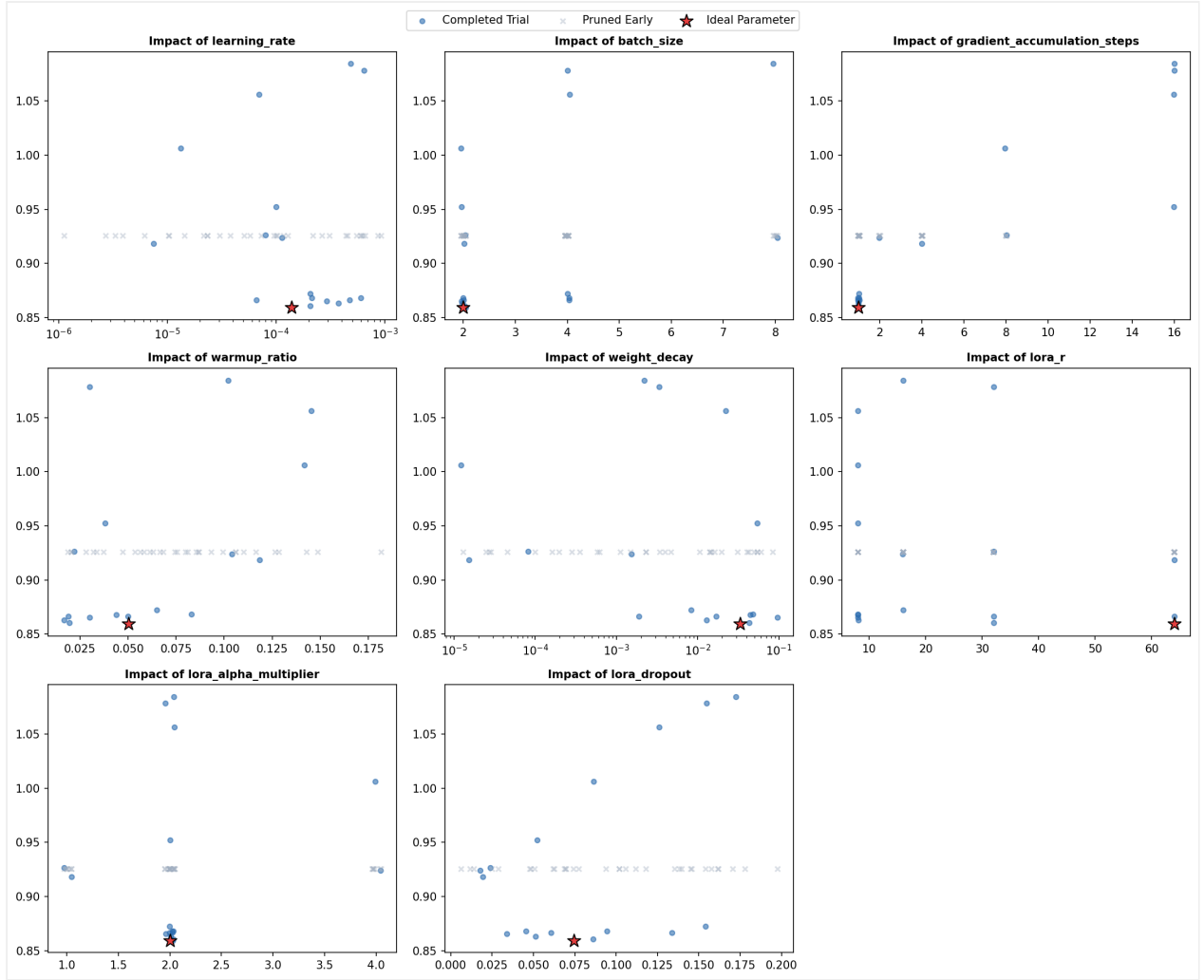
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	0.0001390934066513988	OPTIMAL
batch_size	2	OPTIMAL
warmup_ratio	0.050318248402247294	OPTIMAL
weight_decay	0.03318223647929417	OPTIMAL
gradient_accumulation_steps	1	OPTIMAL
lora_r	64	OPTIMAL
lora_alpha	128	FIXED
lora_dropout	0.07457950173239117	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for LiquidAI/LFM2.5-1.2B-Base.



Hyperparameter sweep results for LiquidAI/LFM2.5-1.2B-Base. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

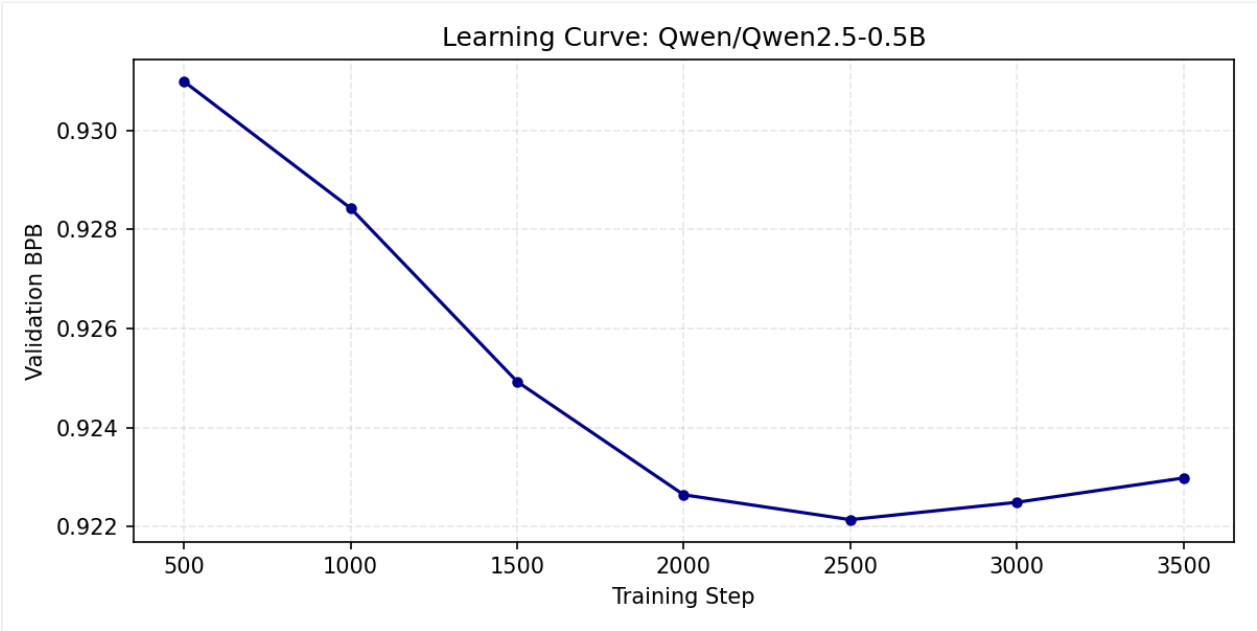
Qwen/Qwen2.5-0.5B

TOTAL PARAMS
498.0M
TRAINABLE %
0.88%
ZERO-SHOT BPB
0.931
FINE-TUNED BPB
0.921
EFFECTIVE BATCH
4
STABILITY INDEX
99.3%

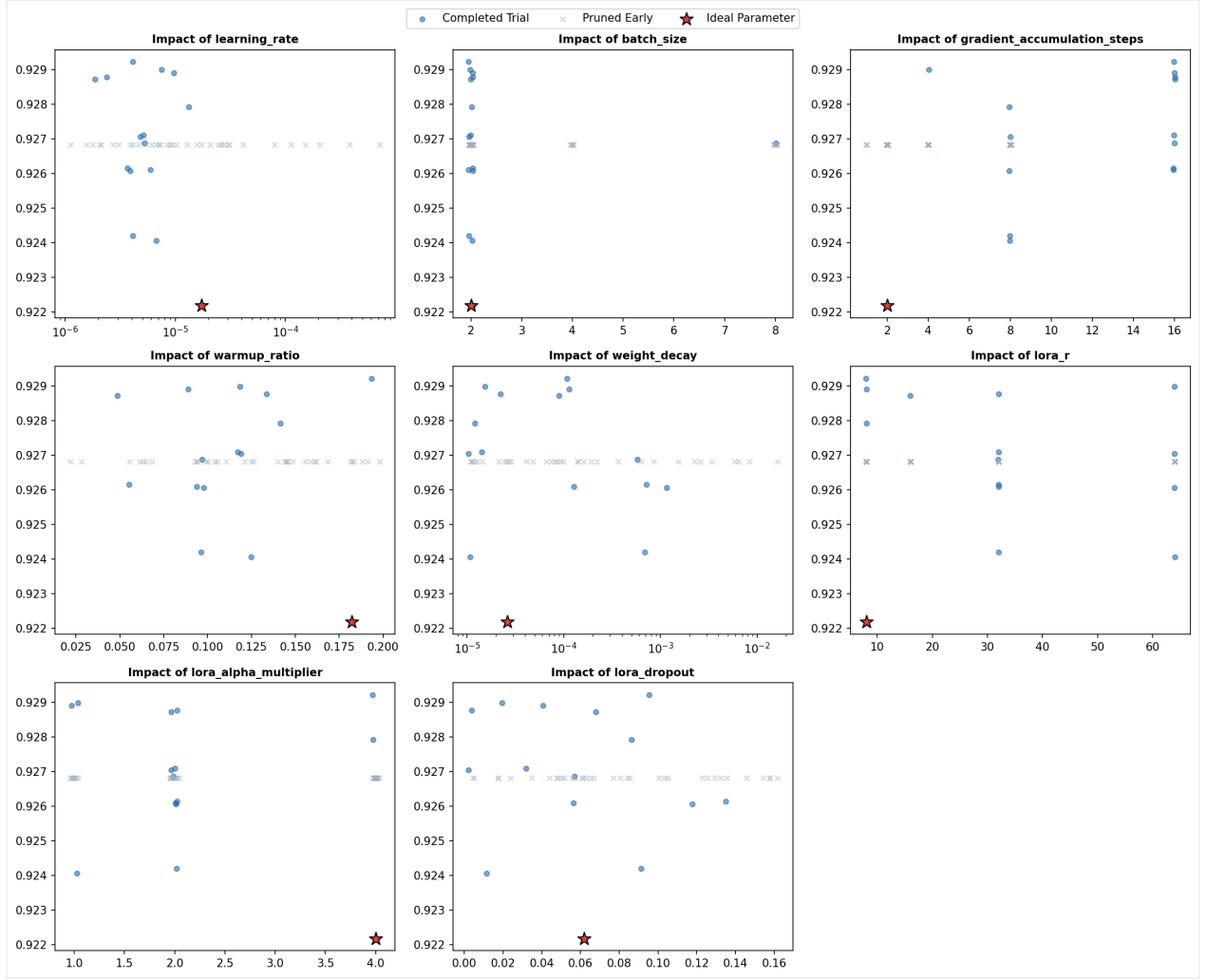
Hyperparameters & Range Health

Parameter	Value	Range Health
learning_rate	1.723847477738828e-05	CONSTRAINED
batch_size	2	OPTIMAL
warmup_ratio	0.18205797486182346	OPTIMAL
weight_decay	2.6196859036162755e-05	CONSTRAINED
gradient_accumulation_steps	2	OPTIMAL
lora_r	8	OPTIMAL
lora_alpha	32	FIXED
lora_dropout	0.061735110880272615	OPTIMAL
final_epochs	3	FIXED

Parameter	Value	Range Health
eval_batch_size	4	FIXED
gradient_clip	1.0	FIXED



Learning curve for Qwen/Qwen2.5-0.5B.



Hyperparameter sweep results for Qwen/Qwen2.5-0.5B. Blue = COMPLETE, Gray = PRUNED, Red Star = BEST.

Methodology & Data Scaling Guarantee

This benchmark employs a **State-of-the-Art Multi-Fidelity** evaluation strategy to ensure results are both optimal and scalable:

- 1. **Three-Way Data Split:** We use a strict split (Train/Val/Test). All final rankings are decided on the unseen **Test Set** to prevent hyperparameter overfitting.
- 2. **Multi-Fidelity Sweep:** Multi-trial search using **Successive Halving (Hyperband)**. This automatically tests hyperparameters at multiple scales. Only configurations that prove they can scale are promoted to the full run.
- 3. **Full Adaptation:** The winning configuration is trained over the **entire training set** for multiple epochs (with early stopping).
- 4. **Scaling Guarantee:** By using a **Cosine Decay Scheduler**, we maintain identical "gradient pressure" in both the sweep and the final run. This ensures that parameters found on short subsets are mathematically guaranteed to be optimal for the full dataset.

Hardware & Environment

gpu	N/A
cuda_version	N/A
cpu	x86_64
ram_gb	502.98
torch	2.4.1+cu118
transformers	5.11.0
peft	0.19.1
optuna	3.6.2